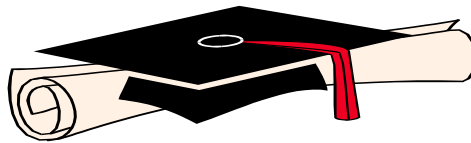




Elementary Programming with C

Semester: 1
Duration: 90 minutes

Curriculum Code: 6629
Total Marks: 20



Practical Examination Paper

Mobile Phone Store Management Program

1. Data Structure

[2.5 mark]

+ Create a data structure with the following specification:

struct Mobile		
Data Type	Information Field	Description
char	name [25]	Mobile's name
char	manufacturer [25]	Manufacturer
int	release_year	Release year
float	price	Mobile's price

2. Menu-based Program

[2.5 mark]

+ Create a menu based program as follows:

```
+-----+
|           MOBILE PHONE STORE MANAGEMENT PROGRAM           |
+-----+
| 1. Input | 2. Sort | 3. Analyze | 4. Find | 5. Save | 6. Open | 7. Exit |
+-----+
```

Note:

+The menu is displayed when the program starts.

+After each choice of the user is completed, ask him/her:

```
Do you want to continue ?
- Yes, I do. (press 'y', 'Y')
- No, I don't. (press 'n', 'N')
- Please clear the screen ! (press 'c', 'C')
Your choice:
```

+If the user enters:

- 'y' or 'Y': repeat the menu.
- 'n', 'N', or '7': exit the program.
- 'c' or 'C': clear the screen.

+Ask the user to re-enter when his/her choice is invalid.

3. Input a list of Mobiles [2.5 mark]

When user selects the option “**1. Input**” from the menu, do the followings:

- Ask the user to enter the total number of **Mobiles**.
- Input information for each **Mobile** in the list.
- Validate data input by the following rules:

Data Anomalies	Message
release_year < 1999	Mobile release year must be greater than or equal to 1999.
price < 500 or price > 6000	Mobile price must be between 500(\$) and 6000(\$).

- If the user enters invalid data, ask him/her to re-enter.

Note: This module must be coded in a function whose parameter is a pointer to the struct **Mobile**.

Input Example: (texts in **bold** are input by user at runtime)

```
Please input Mobile[1]:  
Name: iPhone 6 Plus  
Manufacturer: Apple  
Release Year: 2014  
Price: 699
```

4. Sort Mobiles by name in ascending order [2.5 mark]

When user selects the option “**2. Sort**” from the the menu:

- Sort Mobiles by name in ascending order (from A-Z).
- Displays details of each Mobile in the list after the above re-arrangement.
For example:

Smartphone	Manufacturer	Release Year	Price(\$)
iPhone 6 Plus	Apple	2014	699
iPhone SE	Apple	2016	599
Nexus	Google	2016	599

Note: This module must be coded in a function whose parameter is a pointer to the struct **Mobile**.

5. Analyze statistics of Mobiles by manufacturer [2.5 mark]

When user selects the option “3. Analyze” from the menu :

- Calculate how many Mobiles are there from a specific manufacturer.
- Display the statistics results. For example:

```
Statistics Result:
+ There are 2 Mobile phone(s) manufactured by Apple.
+ There are 1 Mobile phone(s) manufactured by Google.
```

Note: This module must be coded in a function whose parameter is a pointer to the struct **Mobile**.

6. Find Mobiles by manufacturer and price [2.5 mark]

When user selects the option “4. Find” from menu:

- Ask the user to input a *manufacturer*, a *min* and a *max* price.
- Display all the Mobiles that are from the *manufacturer* (input by user) with prices between *min* and *max*. For example:

```
Manufacturer: Apple
Min price($): 300
Max price($): 700

Search Results:
+-----+
| Smartphone | Manufacturer | Release Year | Price($) |
+-----+
| iPhone 6 Plus | Apple       | 2014        | 699      |
+-----+
| iPhone SE    | Apple       | 2016        | 599      |
+-----+
2 smartphones found.
```

Note: This module must be coded in a function whose parameter is a pointer to the struct **Mobile**.

7. Save to file

[2.5 mark]

When user selects the option “5. Save” from the menu, do the followings:

- Ask the user to enter a file name.
- Create the file in binary mode.
- If the program cannot open file, notify the user about that error and return to end this function.
- Write data of the list into the file.
- Notify the user that data has been saved successfully.
- Close the file.

```
Your Choice: 5
Save the list into file:
File name: C:\Xampp\htdocs\Mobile.dat
Data successfully saved into file!
```

Note: This module must be coded in a function whose parameter is a pointer to the struct **Mobile**.

8. Open File and display the list

[2.5 mark]

When user selects the option “6. Open” from the menu, do the followings:

- Ask the user to enter a file name.
- Open the file in binary mode.
- If the program cannot open file, notify the user about that error and return to end this function.
- Read data out of the file, fill into the list and display them all.
- Notify the user that data has been saved successfully.
- Close the file.

```

Your Choice: 6
      Open file and display the list:
File name: C:\Xampp\htdocs\Mobile.dat
+-----+
|Smartphone |Manufacturer |Release Year |Price($) |
+-----+
|iPhone 6 Plus |Apple       |2014        |699      |
+-----+
|iPhone SE     |Apple       |2016        |599      |
+-----+
|Nexus         |Google      |2016        |599      |
+-----+

```

Note: The user must input the list of Mobiles before selecting any other options from the menu, except "Clear screen" and "Exit".

Vocabulary

declare	Khai báo	parameter	Đối số, tham số
follows , following	Như sau	structure array	Mảng cấu trúc
arrange	Sắp xếp	structure pointer	Con trỏ cấu trúc
manage	Quản lý	store, save	Lưu trữ, lưu
Mobile	Điện thoại thông minh	format	Định dạng
manufacturer	Nhà sản xuất	module	Phần, câu, mô đun
price	giá	information	Thông tin
release year	Năm phát hành	include	Bao gồm
calculate	Tính toán	create	Tạo
ascending order	Thứ tự tăng dần	include	Bao gồm
fail	Thất bại	analyze statistics	Phân tích thống kê

validate	Xác thực	invalid	Không hợp lệ
notify	Thông báo		